Adopted Levels

Type	Author	History Citation	Literature Cutoff Date
Full Evaluation	C. Morse	NDS 182, 130 (2022).	14-Sep-2021

Q(β^-)=−4409 *SY*; S(n)=5855 *SY*; S(p)=2837 *SY*; Q(α)=1.089×10^4 *9* 2021Wa16
ΔQ(β^-)=863, ΔS(n)=833, ΔS(p)=859 (2021WA16).
S(2n)=13356 SY *801*, S(2p)=4800 SY *806*, Q(εp)=1224 SY *938* (2021WA16).
Experimental Q(α)=1.089E+4 *7* (2006Og05).

^{291}Lv has been produced in two experiments at JINR through the ^{245}Cm(^{48}Ca,2n) reaction using the U400 cyclotron coupled to DGFRS (2004Og07,2006Og05). An additional event that was tentatively identified as originating from ^{293}Lv in 2012HO12 was later reassigned to ^{291}Lv in 2016HO09. Four events have been observed in total. Candidate events were selected by the observation of chains of time- and position-correlated α decays, initiated with the implantation of evaporation residues in a position-sensitive focal-plane detector, and terminated by spontaneous fission. Positive identification of ^{291}Lv was based on comparison of the known properties of the daughter nuclei to those of the observed decay chains.

Half-lives, branching ratios, and α-decay energies in this evaluation have been computed from the individual events listed in the references above. Half-life uncertainties have been computed according to the method of 1984SC13. An additional 10 keV systematic uncertainty is assumed for the α-decay energies, which is added in quadrature to the averaged statistical uncertainty.

^{291}Lv Levels

E(level)	$T_{1/2}$	Comments
0	19 ms *+17−6*	%α=100; %SF≤20 E(level): Assumed ground state. $T_{1/2}$: From four events.